# Benjamin Chou

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## PROFESSIONAL SUMMARY

3rd-year ECE PhD Student at Purdue University specializing in AI/ML research for audio, music, and multimodal systems. Researching generative model steering, stem separation, Transformers for error detection, and training-free strategies for Multimodal LLMs. I am proficient in Audio DSP, Python, PyTorch, TensorFlow, and HPC clusters. I also have organizational and communication skills (experience organizing 3 AI4Music workshops).

## EDUCATION

**Ph.D. in Electrical & Computer Engineering** | *Purdue University*, West Lafayette, IN — 2023 - Present
- GPA: 4.0/4.0

**B.Sc. in Electrical Engineering** | *National Cheng Kung University*, Tainan, Taiwan — 2020 - 2023
- GPA: 4.0/4.3

## EXPERIENCE

**PhD Resident (Incoming)** | *Google X*, Mountain View, CA — Apr '26 – Jun '26

**Audio Applied Research Scientist Intern (Incoming)** | *Shure*, Niles, IL — Jul '26 – Sep '26

**Applied Research Scientist Intern** | *Reality Defender*, Manhattan, New York — May '25 – Sep '25
- Developed an in-context learning framework with large audio language models, achieving SOTA deepfake detection on in-the-wild speech (+54.9% avg Macro F1 improvement) (Natural Language Processing, RAG, ICL, Audio LLMs, and model routing).

**Graduate Research Assistant** | *Under Dr. Yung-Hsiang Lu, Purdue University*, West Lafayette, IN — Jan '24 – Present
- Developed an AI tutoring model with 6× higher F1 (AAAI 2025) and LadderSym with 2× better missed-note detection (ICLR 2026) (PyTorch, NLP, Audio Transformers).
- Leveraged HPC clusters and Linux environments to engineer generative pipelines for music mistakes, producing two datasets.
- Organized 'AI for Music' workshops at AAAI 2025, IEEE ICME 2025, and NeurIPS 2025, showcasing leadership and communication.

**Startup Founder** | *LocaLens*, West Lafayette, IN — Mar '23 – Jan '24
- Launched an app for menu translation, currency conversion, and allergen alerts using Flutter and Android Studio.
- Built an MVP in 4 months with weekly sprints, emphasizing leadership and collaboration.
- Integrated Google ML Toolkit for cross-platform AI with computer vision and real-time deployment.

**Research Assistant** | *Under Dr. Hsun-Ping Hsieh, National Cheng Kung University*, Taiwan — Mar '23 – Aug 2023
- Raised cancer-screening invitation accuracy to 83% using scikit-learn and tree-based classification (machine learning, statistics, data science).
- Elevated flash-chip yield by 17% with ensembling and Gaussian processes (NumPy, pandas).

## PROJECTS

**Improving Developer Code Understanding with GitHub Issues and Retrieval-Augmented Generation** | *ChatGPT API, REST, Git*
- Built a RAG-based doc tool with GitHub issues; raised README clarity by 116% and code clarity by 16%.

**Multi-Agent Self-Play for Beating Atari Games** | *OpenAI Gym, PettingZoo, PyTorch, RL, Multi-agent systems*
- Used DQN/PPO self-play to improve Atari win rates by 11–32%.

**Spectral Image Inpainting with Deep Learning** | *MATLAB, C++, CUDA*
- Recovered 98% corrupted hyperspectral data at 0.02 RMSE using convex optimization and deep learning.

## HONORS & AWARDS

- Outstanding Student Scholarship, awarded to top 2 students in NCKU-Purdue program (2020–2023).
- AAAI Student Scholarship (2025), selected for NeurIPS volunteer (2025).

## PUBLICATIONS

- **B. Chou**, P. Jajal, N. J. Eliopoulos, J. C. Davis, G. K. Thiravathukal, K. Y.-J. Yun, Y.-H. Lu, "LadderSym: A Multimodal Interleaved Transformer for Music Practice Error Detection", *ICLR 2026 (top 3.85%)* [Link]
- **B. Chou**, et al., "Enabling Note-Wise Music Separation and Editing via a Partially Synthetic Data Engine", *In Progress*
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y. Lu, "AdaPerceiver: Transformers with Adaptive Width, Depth, and Tokens", *CVPR 2026 Findings* [Link]
- **B. Chou**, Y. Zhu, S. Koppisetti, "ICLAD: In-Context Learning with Comparison-Guidance for Audio Deepfake Detection", *Under Submission*
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y. Lu, "Inference-Time Alignment of Diffusion Models with Evolutionary Algorithms", *CVPR 2026 Findings* [Link]
- **B. Chou**, P. Jajal, N. J. Eliopoulos, T. Nadolsky, C.-Y. Yang, N. Ravi, J. C. Davis, K. Y.-J. Yun, Y. Lu, "Detecting Performance Errors with Transformers", *AAAI 2025* [Link]
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y.-H. Lu, "Token Turing Machines are Efficient Vision Models", *WACV 2025* [Link]

## PATENT PENDING

- System and Method for Detecting Musical Performance Errors, Patent Application 2025.
- ICLAD: In-Context Learning with Comparison-Guidance for Audio Deepfake Detection, Provisional Patent Application 2025.